The Social Integration of International Migrants

Evidence from the Networks of Syrians in Germany

Bailey, Johnston, Koenen, Kuchler, Russel, & Stroebel (2025)
Presented by Yuxuan He

February 6, 2026

UC3M

Roadmap

1. Context & Data Infrastructure
2. Place Effects vs. Sorting
3. Mechanisms: Frictions vs. Volume
4. Policy Effectiveness
5. Economic Consequences

Conceptual Framework (AI Visualization)

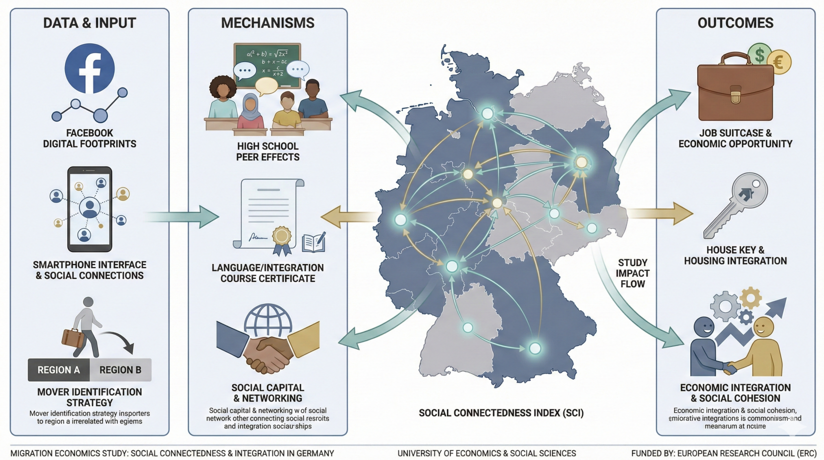

Figure 1: Visual Abstract: From Migration to Integration

Roadmap

1. Context & Data Infrastructure
2. Place Effects vs. Sorting
3. Mechanisms: Frictions vs. Volume
4. Policy Effectiveness
5. Economic Consequences

Motivation: The Integration Puzzle

The 2015 Refugee Crisis:

- Germany received $>$ 1 million refugees (mostly Syrian).
- *Policy shift:* From "Crisis Management" to "Long-term Integration."

The Knowledge Gap:

- We have administrative data on *residential* location.
- We lack granular data on *social* interaction.
- **This Paper:** Uses Facebook data to measure social integration at the district level.

Data Construction: Three Measures of Integration

Sample: $\approx$ 350k Syrian Migrants & 18m German Natives.

1. **Number of Native Friends:**
   - The number of native German friends a Syrian migrant user has in the same or a bordering county
2. **Language Integration:**
   - Percentage of Facebook content produced in **German**.
3. **Group Integration:**
   - Participation in local German Facebook groups (e.g., sports clubs).

**Note: These measures are validated against IAB-BAMF-SOEP survey data.*

Integration Trends Over Time

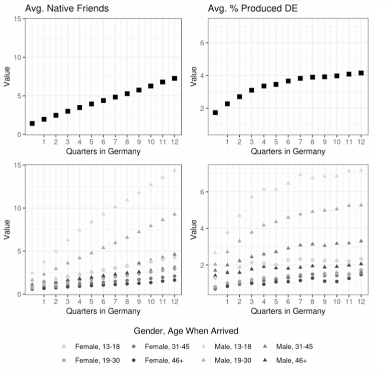

Figure 2: Integration measures ramp up with time since arrival

Migrants become increasingly socially integrated as they spend more time in Germany

Data Validation: Is it Real?

Skepticism: Are Facebook friends just "cheap talk"?

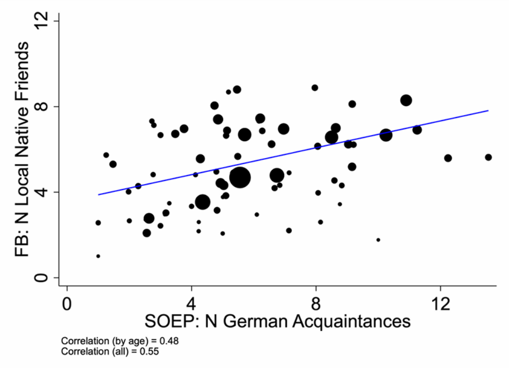

Figure 3: Validation: Facebook Measure vs. SOEP Survey Data

Finding: Strong correlation ($R^2 > 0.6$) between the Facebook measure and survey responses on "time spent with Germans."

Roadmap

1. Context & Data Infrastructure
2. Place Effects vs. Sorting
3. Mechanisms: Frictions vs. Volume
4. Policy Effectiveness
5. Economic Consequences

Stylized Fact: The Geography of Integration

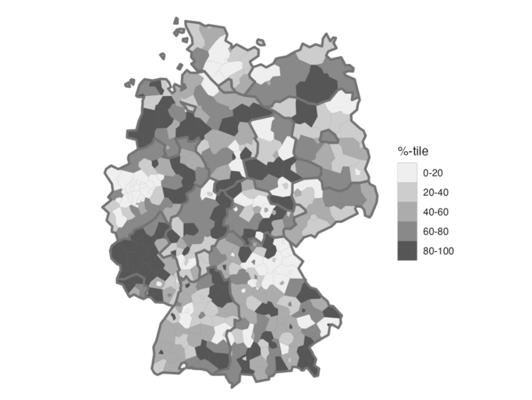

Figure 4: Regional Heterogeneity in Social Integration

Key Patterns: 1. **East-West Divide:** Significantly lower integration in former East Germany.
2. **Urban Paradox:** Lower integration in major cities despite high migrant presence.

Identification: The Mover's Design

Challenge: Is high integration in Region A due to *Sorting* (people) or *Treatment* (place)?

Strategy: Track migrants moving between regions.

$$y_{it} = \alpha_i + \gamma_{r(i,t)} + \tau_t + \varepsilon_{it}$$

- We test if integration converges to the destination level after a move.
- α_i: Individual Fixed Effect.
- γ_r: Place Fixed Effect.

Evidence: The Impact of Moving (Event Study)

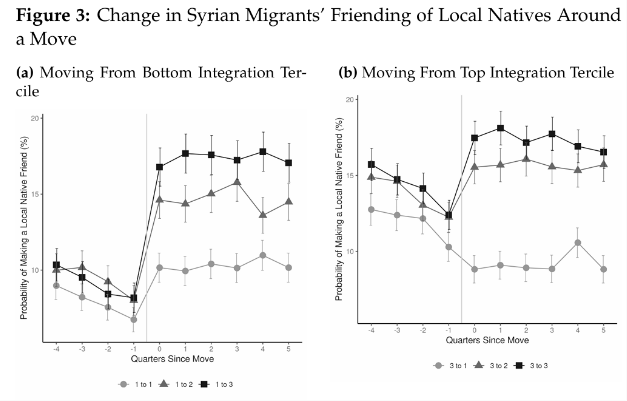

Figure 5: Event Study of Migrant Movers

Visual Proof: Migrants moving to better areas see immediate improvement. Flat pre-trends confirm validity.

Decomposition Results

Quantitative Breakdown of Variance:

- **Place Effects (γ_r):** Explain **75%** of the cross-regional variation.
- **Sorting:** Explains only **25%**.

Conclusion: Local institutions matter more than individual sorting.

Roadmap

1. Context & Data Infrastructure
2. Place Effects vs. Sorting
3. Mechanisms: Frictions vs. Volume
4. Policy Effectiveness
5. Economic Consequences

Decomposing the Place Effect

We split the "Friending Rate" into two multiplicative components:

$$N(\text{Friend})_{SY-DE} = \underbrace{N(\text{Friend})_{DE-DE}}_{\text{General Friendliness}} \times \underbrace{\frac{N(\text{Friend})_{SY-DE}}{N(\text{Friend})_{DE-DE}}}_{\text{Relative Friending}}$$

- **General Friendliness :** Probability that natives befriend other natives (Scale).
- **Relative Friending :** Probability natives befriend migrants *relative* to natives (Bias/Friction).

What Drives the Variation? (Scatter Plot)

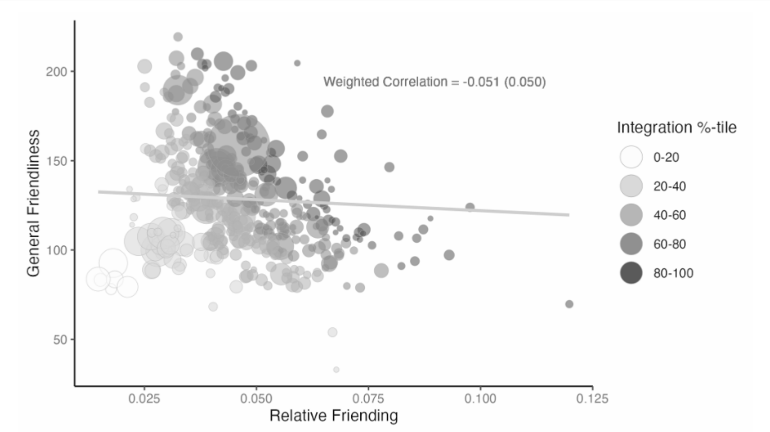

Figure 6: Relative Friending (X) vs. General Friendliness (Y)

Conclusion: Variation in Relative Friending (Bias) explains 65% of total variance.

Are Biases Fixed? (Native Movers)

Do natives have immutable preferences (e.g., racism)?

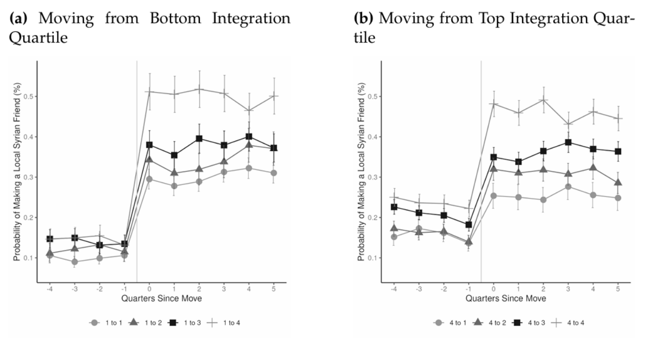

Figure 7: Native Movers Event Study

Result: When Germans move to high-integration areas, they also begin to befriend more migrants. The "bias" is an **institutional equilibrium**, not a fixed trait.

Roadmap

1. Context & Data Infrastructure
2. Place Effects vs. Sorting
3. Mechanisms: Frictions vs. Volume
4. Policy Effectiveness
5. Economic Consequences

Policy 1: Integration Courses

Design: Instrumental Variable (IV) Strategy.

- *Instrument (Z_r):* Supply of certified teachers in 2014 (Pre-Crisis).

	Integration	General Friendliness	Relative Friending	Language	Employ. / Training
Log Integration Courses per Syrian	1.792***	0.154	1.542***	0.342***	0.902***
	(0.34)	(0.17)	(0.27)	(0.07)	(0.15)
Control Covariates	x	x	x	x	x
Control Log General Unemployment Rate	x	x	x	x	x
N	390	390	390	390	384

Figure 8: IV Estimates of Course Effects

Result: Column 1 shows a statistically significant positive effect. 10% more courses → 18% higher relative friending.

Policy 2: The Contact Hypothesis

Design: Exploits quasi-random variation in the number of Syrian refugees in High School cohorts.

	Syrian Friends		Syrian Friends (Excluding Classmates)		Syrian Friends (Excluding Syrian Classmates and their Friends)	
Syrian in Cohort	0.020***	0.020***	0.005***	0.005***	0.005***	0.005***
	(0.002)	(0.002)	(0.002)	(0.002)	(0.001)	(0.001)
Syrian in Cohort x Standardized Cohort Size		-0.007***		-0.003***		-0.003***
		(0.001)		(0.001)		(0.001)
School FE	X	X	X	X	X	X
Birth Year x County FE	X	X	X	X	X	X
N	115,625	115,625	115,625	115,625	115,625	115,625
Mean in Control Cohort	0.054	0.054	0.029	0.029	0.027	0.027

Figure 9: Impact of High School Exposure on Future Friending

Result: Exposure to Syrian peers in HS significantly increases the probability of befriending Syrians later in life.

Roadmap

1. Context & Data Infrastructure
2. Place Effects vs. Sorting
3. Mechanisms: Frictions vs. Volume
4. Policy Effectiveness
5. Economic Consequences

Instrumental Value of Social Capital

Social integration is not just about "being nice"—it has tangible economic returns.

	Native Friend Helped Find Job		Native Friend Helped Find Housing		Native Friend Helped with School		Native Friend Helped with Bureaucracy		Native Friend Helped with Healthcare	
N Local Native Friends	0.497***	0.347***	0.370***	0.375***	0.159*	0.191*	0.444***	0.375***	0.131	0.084
	(0.095)	(0.109)	(0.096)	(0.121)	(0.084)	(0.102)	(0.095)	(0.109)	(0.090)	(0.112)
Control Covariates		X		X		X		X		X
N	2,738	2,687	2,738	2,687	2,738	2,687	2,738	2,687	2,738	2,687
Sample Mean	48.32	48.49	47.59	47.45	26	25.98	54.67	54.45	32.58	32.45

Figure 10: Correlation between Friends and Economic Help

Migrants with more German friends are significantly more likely to receive help finding jobs and housing.

Summary of Main Findings

1. **Heterogeneity:** Migrants' social integration varies across regions, driven by causal, place-based effects (75%).
2. **Mechanism:** This variation reflects local institutions (local equilibrium) rather than immutable native characteristics. The main barrier is "Relative Friending."
3. **Policy 1:** Integration courses causally boost inter-group friending rates.
4. **Policy 2:** Quasi-random exposure in high school increases future friending (Contact Hypothesis).
5. **Outcomes:** Social integration facilitates economic help (jobs, housing).

Thank You!